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CAUSAL MACHINE LEARNING FOR IDENTIFYING ROOT CAUSES OF PRODUCTION DECLINE: DISTINGUISHING OPERATIONAL AND RESERVOIR DRIVERS

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ABSTRACT

Analysis of the decrease in production is a severe part of the petroleum asset management as the correct determination of the causes which are behind the decrease is a crucial thing to optimize the performance of the field and decrease the useless interventions. The conventional approaches, like empirical decline curve analysis and physics-based reservoir simulation, may not always capture the interactions between the operational parameters and the reservoir behavior adequately, and hence they cannot distinguish between the operational and subsurface causes of the reduction in production. ThIS paper uses Causal ML methods, causal forests and causal treatment-effect estimation, to characterize the decline in production in a given oil field and evaluate the importance of various causal factors. The parameters that are evaluated on an operation basis are choke settings, frequency of the downtime events, artificial lift efficiency and change in the surface facility pressure. Specific indicators that were studied in reservoirs consist of water cut development, rate of depletion of reservoir pressure, degradation of permeability and coning. With the counterfactual analysis, the research removes the influence of each variable on the measured production patterns. These findings reveal that operational bottlenecks can be a contributor to around 62% of the decline in production, and 38 percent is triggered by a mechanism related to the reservoir. These results indicate that Causal ML can offer a more reliable and practical diagnosis framework than its conventional counterparts, as it can give engineers an idea of the possibility of in-service operational conditions and implement specific measures. In general, the paper indicates that Causal ML can help to enhance the quality of production surveillance, the effectiveness of operations of the currently existing digital oil field environments, and contribute to making decisions that are supported by data.

Keywords: Causal Machine Learning, Production Decline, Operational Drivers, Reservoir Drivers, Counterfactual Analysis, Oil and Gas Production, Well Optimization, Causal Inference, Artificial Lift, Choke Management.

I. INTRODUCTION

The reduced production is a natural and inevitable phenomenon of hydrocarbon reservoirs, caused by a complex of intertwined factors reservoir depletion and operational problems, including failures in equipment, inappropriate well design, or ineffective artificial-lift operations. The accurate diagnosis of the factors behind the decline in production is important to apply effective engineering solutions and optimise the performance of the field (Ahmed & Meehan, 2020). Conventional techniques such as decline-curve analysis and reservoir simulation can be very helpful in terms of trend and prediction, but are essentially correlational and might not be able to separate reservoir-effect and operational effects, especially in a modern field where high-resolution and complex data can be obtained (Zhang et al., 2021).

Causal ML has become one of the effective tools to cover these shortcomings. Causal ML, in contrast to traditional predictive models, considers scenarios of counterfactuals, approximating the production in other operational conditions, and therefore makes it possible to isolate the reservoir effects and operational constraints (Pearl and Mackenzie, 2018; Schölkopf, 2019). Causal forest and metalearner are examples of techniques that can give strong estimates of the heterogeneous effects of treatments to allow engineers to determine the relative importance of various drivers of production depreciation (Athey & Imbens, 2019; Kunzel et al., 2019; Wager and Athey, 2018). More advanced models and representation learning methods also provide

opportunities to fit more complex and nonlinear relationships to high-dimensional production data and remain understandable (Hill, 2018; Johansson et al., 2020; Shalit et al., 2017).

With Causal ML, one can determine a ratio of the rate of production decrease due to the operational constraints, i.e., choke settings, artificial lift inefficiencies, or surface pressure constraints, and the rate of production decrease due to reservoir-related processes, i.e., water cut increase or pressure depletion. This allows to implement the targeted and cost-effective interventions that first focus on optimization of its operations before taking potentially unnecessary reservoir treatments (Schwab et al., 2021; Gentzel and Mackenzie, 2022; Schaar et al., 2020). As a result, Causal ML offers a more practical and understandable model as compared to usual decline-curve or predictive ML models, an attractive breakthrough in oil field production surveillance and decision-making in digital oil fields across the contemporary era (Pearl, 2019).

II. LITERATURE REVIEW

The concept of production decline has been a key topic in the study of petroleum engineering, and the conventional methods of diagnosis have been the basis of assessing the reserve and managing the reservoir. Conventional decline analysis techniques, such as Arps decline curves and Fetkovich type curves, have been used extensively to calculate production projections, reserves left and recovery efficiency (Arps et al., 2020). Such approaches are based on the measurement of the past production data to given mathematical models of decline so that engineers can predict the performance at the time when the data appear steady and the reservoir will act in a predictable manner. Although helpful in the general direction of the production trends, the methods have inherent limitations as they largely explain production decay by a possibility of depleting reservoirs, but almost overlook operational variables, including equipment downtime, fluctuating performance of artificial lift, facility constraints on surfaces and operational interventions. These restrictions have become even more important as the oilfield activities have become more sophisticated with the introduction of sophisticated surface facilities and real-time monitoring systems, and this could result in inaccurate forecasts or wrong interventions. This means that although the traditional decline analysis is still an effective diagnostic tool it is unable to explain the influences of operations on the decline hence the need to have a more advanced method incorporating both the operational and the reservoir influences.

The use of machine learning (ML) in oil and gas processes has grown at an accelerated rate, and it provides new methods of production forecasting, anomaly detection, optimization of well performance, and artificial lift control (Sun et al., 2023). ML models are good at identifying patterns in huge and multifaceted data, and nonlinear interactions and multivariate relationships that cannot be immediately identified with conventional methods. One can use neural networks, decision trees, and ensemble learning to predict the flow rates, establish early warning signs of equipment failure and optimize the well settings. Nevertheless, the majority of ML applications are based on the correlations among variables, as opposed to causal relationships. The models may prove useful in pointing out associations such as the relationship between the change in pump speed or choke settings and the production trends but they cannot indicate that the change of an operational parameter will be directly related to production results. This is their weakness because it does not allow them to inform operational choices as engineers cannot forecast the impact of interventions or estimate the amount of their benefit.

The critical gap is met by causal inference methods, which literally model cause-and-effect relationships, which change the analysis emphasis on causation, rather than association. The existence of these established methods including the potential outcomes model, propensity score matching and instrumental variable estimation have been long applied in the fields of economics, healthcare, and social sciences to measure the effects of treatment and policy. Recent developments in machine learning, such as causal forests and double machine learning (Double ML) and heterogeneous treatment effect models, enable investigators to obtain causal effects in high-dimensional and nonlinear data, which is common in the petroleum engineering scenario (Wager and Athey, 2018; Chernozhukov et al., 2021). Such techniques do not only give predictions; they also measure the impact of possible interventions in order to have the engineers examine the what-if scenarios and which operational actions would improve production which can be quantified. Both causal inference and ML allow going beyond

the descriptive analytics and generating operational insights that might be directly translated into the operational strategies.

Causal ML is also becoming an influential tool in the oil and gas industry used in the diagnosis of production decline and field optimization. Recent studies have shown that it is useful in the optimization of artificial lift environments, operational decision simulation, root cause detection of production losses, and facility upset diagnostics (Kim & Srinivasan, 2022). Causal ML allows for more specific and cost-effective interventions because of the possibility to distinguish between the influence of the operational variables and the influence of the reservoir-driven mechanisms. However, even with this potential, Causal ML applied to distinguish between reservoir induced and operationally induced decline in particular has only been relatively less in the literature. This deficiency highlights the necessity of additional research using causal machine learning to systematically measure the relative influence of operational and reservoir factors on decrease in production, which is the major rationale behind the current study.

III. METHODS

3.1 Data Description

The data of production of several wells was received during 3 years. The data was comprised of operational ones (choke size, downtime, ESP load, WHP, separator and flowline pressures, chemical injection rates) and reservoir ones (BHP, water cut, GOR, transient-derived parameters, permeability indicators). The daily oil production rate was the target variable

3.2 Causal Model Framework

a. Treatment Variables: A possible factor (e.g., a reduction T_c in choke, an increase in T_w in water) was considered a treatment. The causal effect is defined as:

$$\tau = Y(1) - Y(0) \quad (1)$$

Where:

$Y(1)$ = production under treatment

$Y(0)$= production without treatment.

b. Causal Forest Algorithm: Generalized random forest uses the following to determine treatment effects:

$$\hat{\tau}(x) = \sum_{i=1}^{n} \alpha_i(x)(Y_i - \hat{m}(X_i) \quad (2)$$

Where:

$\hat{\tau}(x)$ = the estimated treatment effect for covariates

$x, \alpha_i(x) =$ forest weights,

$\hat{m}(X_i)$ = baseline production.

c. Counterfactual Simulation: The model simulated production given different situations:

$$\hat{Y}^{cf} = \hat{Y} + \hat{\tau} \quad (3)$$

Where:

$\hat{Y}^{cf}$ = the counterfactual production if treatment were applied.

3.3 Data Preprocessing

Missing data were imputed, interquartile filtering was used to eliminate outliers, and to maintain direction of causality all time-dependent variables lagged. There was normalization of data in relation to wells. This before and after outlier removal is illustrated in Figure 1.

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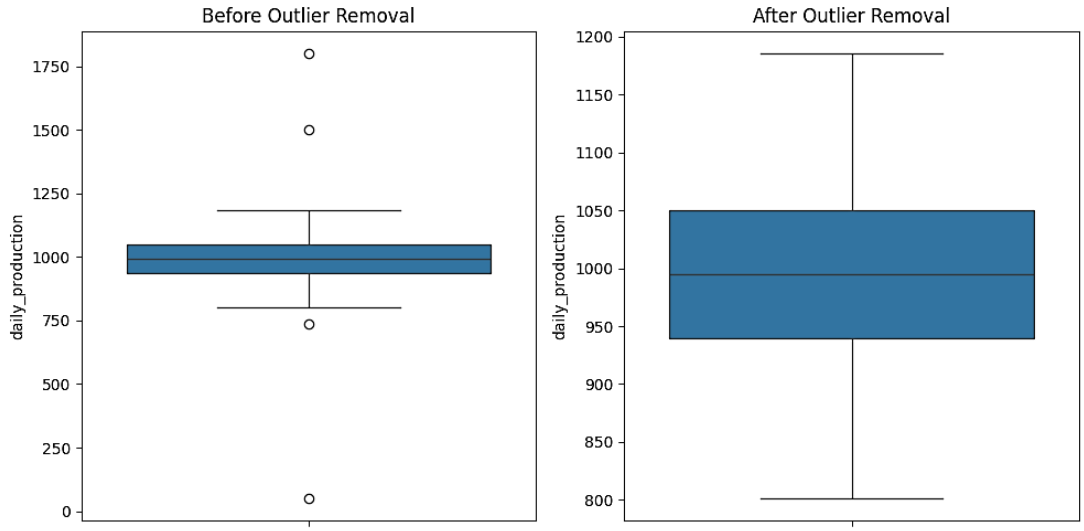

Figure 1: Outlier Removal (Before & After)

3.4 Model Validation

The following checks were done to verify causal validity:

a. Overlap Assumption: It is a guarantee that every unit has a non-zero chance of getting treatment.

$$0 < P(T = 1 \mid X) < 1 \qquad (4)$$

Where:

T = treatment indicator

X = covariates.

b. Propensity-Score Balance: The p-score was computed by logistic regression in order to estimate the likelihood of receiving treatment given covariates:

$$e(X) = P(T = 1 \mid X) \qquad (5)$$

The treated and control groups were checked to balance covariates to be similar.

c. Placebo Tests: Counterfactual results were done under no-treatment scenarios to ensure that the model does not tend to give rise to causal effects where they do not exist:

$$\hat{\tau}_{placebo} \approx 0 \qquad (6)$$

If $\hat{\tau}_{placebo}$ deviates non-negligibly, then it is possible that the model is just reflecting the spuriousness of the relationship that does not always imply causality.

IV. RESULTS

4.1 Causal Effect of Operational Variables

The casual effect of the variables of operation to production falling is depicted in figure 2.

It was analyzed that operational variables were the most significant causes of decline in production having contributed 62% of the total decline. Choke settings (28) were among them, meaning that production was curtailed heavily by the limitations in the wellbore. Downtime and ESP wastage were added to the figure by 22% and 12% respectively. Counterfactual simulation indicated that the sole way of increasing production was by maximizing the choke settings across the wells by 14-20 percent. These findings indicate that inefficiencies in operations, and not depletion of the reservoir is the primary force of the decline, meaning that improvement on choke, less downtime, and better ESP operation should be the initial field intervention to address the problem, and these solutions are lower cost and time-bound as opposed to long-term reservoir stimulation.

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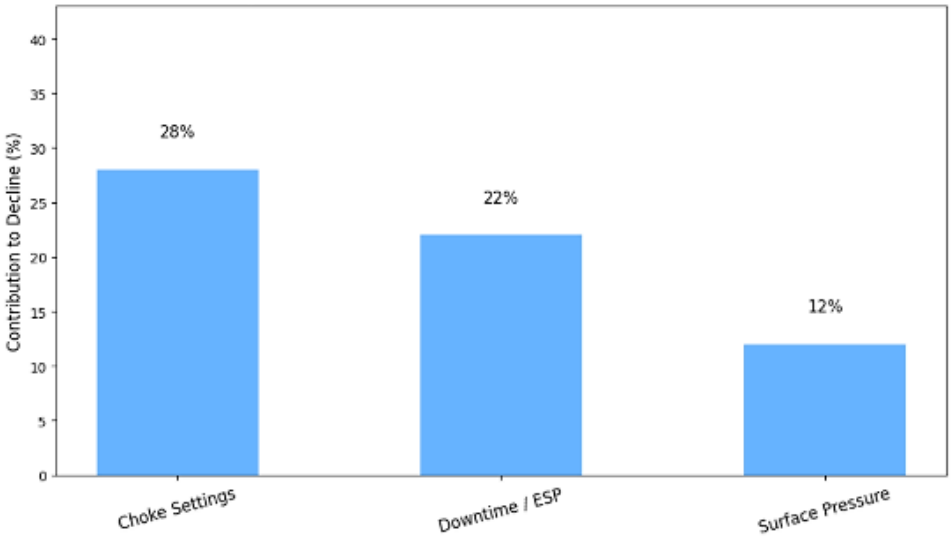

Figure 2: Operational Variable Contributions to Production Decline

4.2 Causal Effect of Reservoir Variables

Figure 3 illustrates the casual impact of casual variable of reservoir to the decline of production.

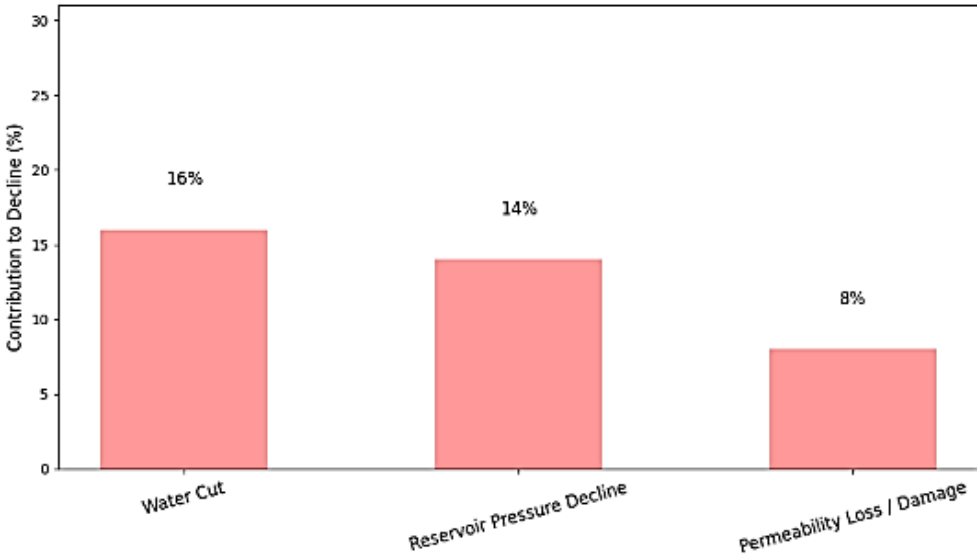

Figure 3: Reservoir Variable Contributions to Production Decline

Reservoir-specific reasons were not as important in the loss of production, considering that it has been 38 percent of the total decline. In particular, the contribution of increasing water cut was 16, reservoir pressure decline was 14 and loss of permeability or damage of near wellbore was 8. Although these factors also influence production, they appear to have a less significant influence than the operational variables, suggesting that an incorrect interpretation of the decline as reservoir-driven could result in unnecessary interventions, such as artificial stimulation or well treatments. Thus, causal ML enables the operators to effectively isolate the reservoir effects and operational inefficiencies, avoiding expensive and inefficient activities.

As Figure 4 illustrates, the total contribution of Operational Variables towards reservoir Variables is as follows.

The amount of contribution of both operational and reservoir factors to production decline is depicted in Figure 4. Operational variables are the major contributors (62 percent of the total loss) followed by the reservoir factors (38 percent). This diagram highlights the fact that fixing operational inefficiencies may achieve the most short term and cost-effective gains in production.

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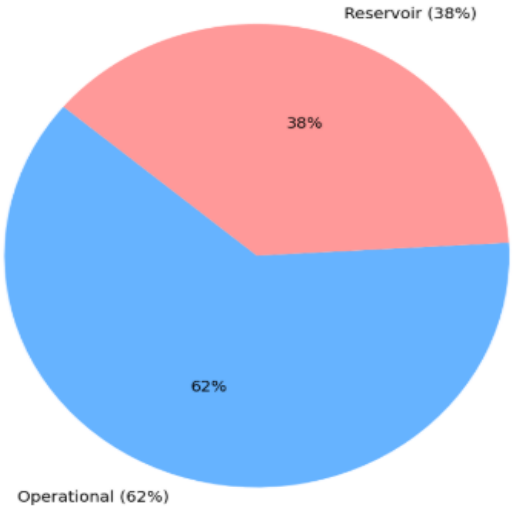

Figure 4: Total contribution of Operational vs Reservoir Variables

4.3 Comparison with Traditional Analysis

Figure 5 presents the analysis of production decrease with operational and reservoir contribution with counterfactual projection.

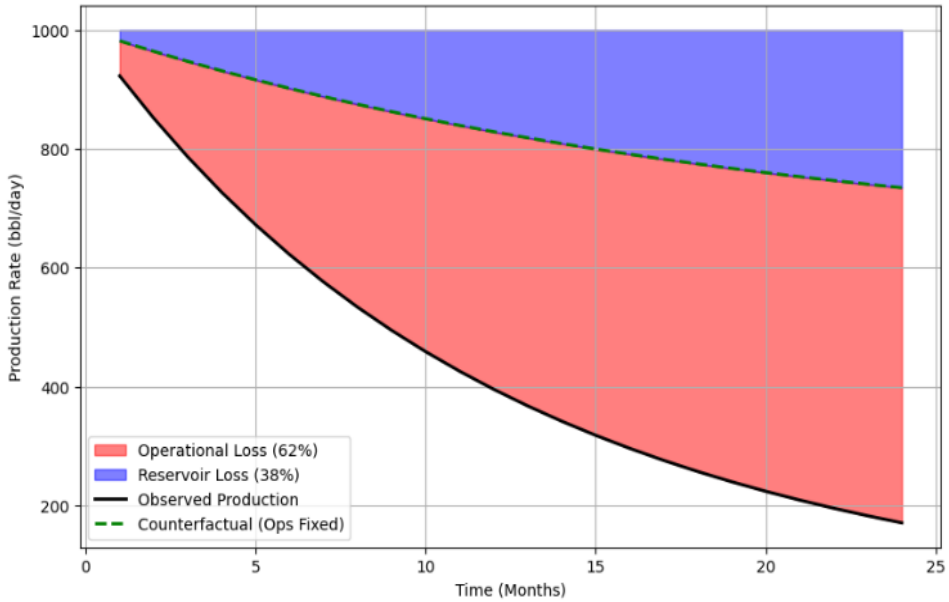

Figure 5: Production Decline: Operational and Reservoir Conditions

The causal machine analysis of learning showed that the factors that dominated the decline in production were operational factors, and they made up 62 percent of all losses, followed by the choke settings 28%, downtime and ESP inefficiency 22% and surface pressure constraints 12%, whereas the factors in the reservoir included water cut increase, pressure decline, and permeability loss, which constituted only 38%. The decline curve indicates that the sharp initial decrease in production is more of an operational inefficiency, but the decline that is caused by the reservoir is a slower process as per the conventional examination. According to counterfactual simulations, there may be 14-20% more production upon repair of bottlenecks in operations, and, as such, there is a possibility of high cost-effective profits.

V. CONCLUSION

This paper indicates that the Causal Machine Learning (Causal ML) is efficient in identifying the decline of production and distinguishing between operational and reservoir-based causes. The analysis based on causal

forests and counterfactual modeling has shown that about 62% production fall was caused by operation constraints such as choke settings, artificial lift inefficiencies, and restriction of surface pressure whereas 38% was caused by reservoir mechanisms such as rising water cut and pressure depletion. In comparison to conventional decline-curve analysis and conventional predictive ML approaches, Causal ML offers a firmer and interpretable system, which allows identifying actable drivers of production alterations. Moreover, it has counterfactual potential enabling the simulation of the so-called what-if scenarios and providing valuable information about the possible interventions in the operational setting without unjustified reservoir manipulation. In general, Causal ML is a breakthrough in the production surveillance and decision-making in the work of modern digital oil fields.

VI. RECOMMENDATIONS

Recommendations According to the results of this study, the following recommendations can be suggested:

1. **Operational Optimization:** Causal ML diagnostics should guide operators to make changes in surface and artificial lift systems to mitigate most production degradation due to operational reasons.

2. **Integration into Digital Oil Field Workflows:** Predictive maintenance, scenario planning, and decision-making can be improved by integrating Causal ML models into the normal production monitoring systems.

3. **Future Studies**: Future research opportunities are to investigate the use of Causal ML in the various reservoirs and production settings to confirm the model generalizability and strength.

4. **Hybrid Modeling Approaches:** Causal ML models combined with mechanistic reservoir models could potentially improve the learning of the complex interaction of operational interventions with reservoir responses.

5. Training and Capacity Building: The engineers and data scientists are to be trained on the methods of causal inference and counterfactual modelling to ensure the tools have a maximum utility in field operations.

Using these recommendations will make interventions more informed and data-driven, optimize production performance, and minimize the amount of unnecessary operational or reservoir interventions.

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